(ii) $$P(X=1) = f(1) = \binom{8}{1}\left(\frac{1}{4}\right)^{1}\left(\frac{3}{4}\right)^{8-1} = 2\left(\frac{3}{4}\right)^{7}$$

(iii) $$P(X \geq 1) = 1 - P(X < 1) = 1 - P(X = 0) = 1 - \left(\frac{3}{4}\right)^{8}$$

Example 11.22

On the average, 20% of the products manufactured by ABC Company are found to be defective. If we select 6 of these products at random and X denotes the number of defective products find the probability that (i) two products are defective (ii) at most one product is defective (iii) at least two products are defective.

Solution

Given that $n = 6$

Probability for selecting a defective product is $\frac{20}{100}$, that is $p = \frac{1}{5}$.

Since X denotes the number defective products, X can take on the values $0,1,2,...,6$

The probability for defective (success) is $p = \frac{1}{5}$ and for failure $q = 1 - p = \frac{4}{5}$, and $n = 6$

Therefore X follows the binomial distribution denoted by $X \sim B\left(6, \frac{1}{5}\right)$.

This gives $$f(x) = \binom{6}{x}\left(\frac{1}{5}\right)^{x}\left(\frac{4}{5}\right)^{6-x}, \qquad x = 0,1,2,...,6.$$

(i) Probability for two defective products is

$$P(X=2) = f(2) = \binom{6}{2}\left(\frac{1}{5}\right)^{2}\left(\frac{4}{5}\right)^{6-2} = 15\left(\frac{4^4}{5^6}\right)$$

(ii) Probability for at most one defective product is

$$P(X \leq 1) = P(X=0) + P(X=1)$$

$$= \binom{6}{0}\left(\frac{1}{5}\right)^{0}\left(\frac{4}{5}\right)^{6-0} + \binom{6}{1}\left(\frac{1}{5}\right)^{1}\left(\frac{4}{5}\right)^{6-1}$$

$$= \left(\frac{4}{5}\right)^{6} + (6)\left(\frac{4^5}{5^6}\right) = 2\left(\frac{4}{5}\right)^{5}$$

Probability for at most one defective product is $2\left(\frac{4}{5}\right)^{5}$.

(iii) Probability for at least two defective products is

$$P(X \geq 2) = 1 - P(X < 2) = 1 - P(X \leq 1) = 1 - 2\left(\frac{4}{5}\right)^{5}$$

Probability for at least two defective products is $1 - 2\left(\frac{4}{5}\right)^{5}$.

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